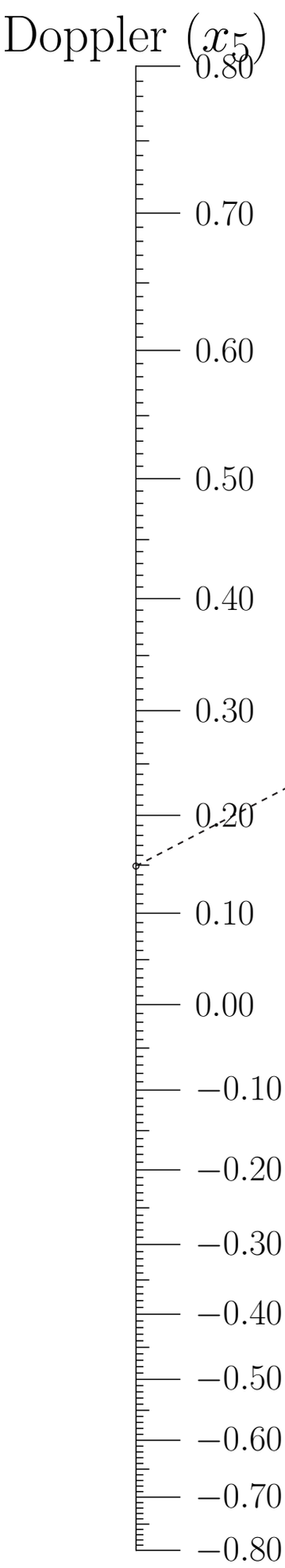
EXACT FORMULA SPECIFICATION

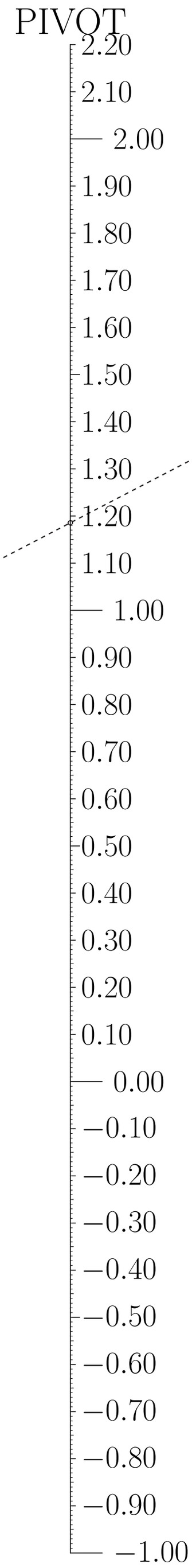
$$L = L_{linear} \times M$$

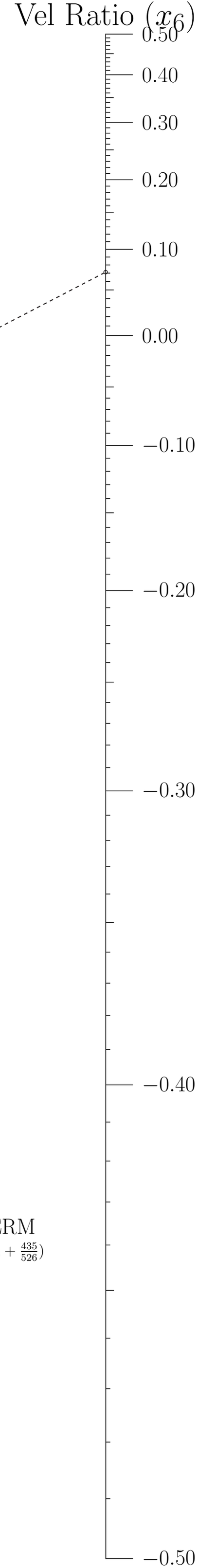
$$M = \exp(\tfrac{345}{509} \cdot x_5) + \frac{x_6}{x_6 + \frac{435}{526}} - \frac{\frac{5}{67}}{x_2} \tanh(x_5 x_8 + \tfrac{2117}{961})$$

Where: $x_5 = UT/L_{lin}, \quad x_6 = U/C_0$

$x_2 = \ln(H/d), \quad x_8 = \ln(T\sqrt{g/d})$







STEP 1: CALCULATE PIVOT TERM

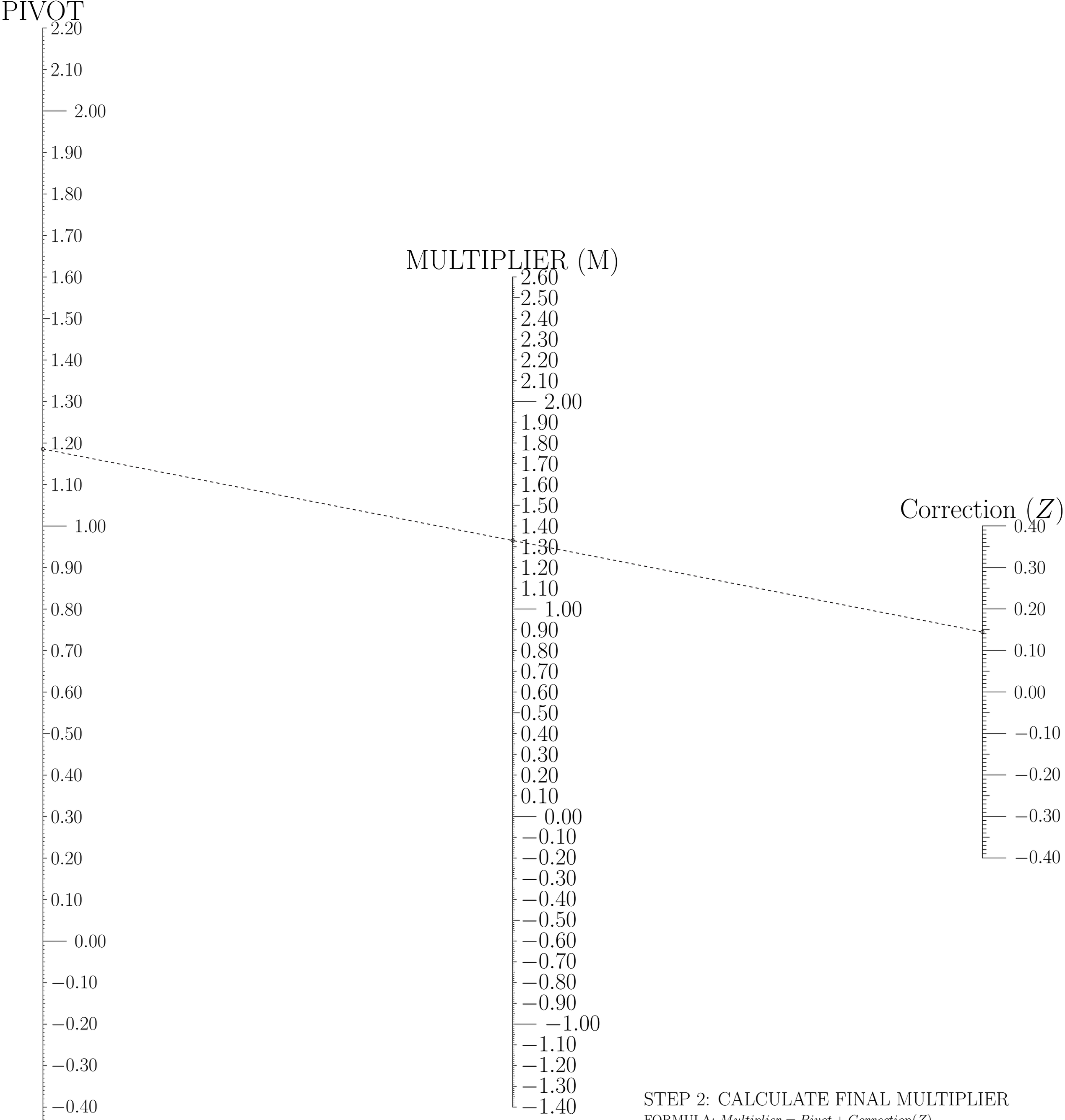
SUB-FORMULA: $Pivot = \exp(\frac{345}{509} \cdot x_5) + x_6/(x_6 + \frac{435}{526})$

INPUT PARAMETERS:

1. Doppler $(x_5) = U \cdot T/L(lin)$
2. Vel Ratio $(x_6) = U/C_0$

INSTRUCTIONS:

A. Draw line from x_5 (Left) to x_6 (Right).
B. Read PIVOT at Center intersection.
C. Write value down for Page 2.



STEP 2: CALCULATE FINAL MULTIPLIER

FORMULA: $Multiplier = Pivot + Correction(Z)$
FORMULA: $Z = -(\frac{5}{67}/x_2) \cdot \tanh(x_5 \cdot x_8 + \frac{2117}{961})$

INPUT PARAMETERS:

1. Rel Height $(x_2) = \ln(H/d)$
2. Dim Period $(x_8) = \ln(T \cdot \sqrt{g/d})$

INSTRUCTIONS:

A. Mark PIVOT (from Page 1) on Left Axis.
B. Calculate Z and mark on Right Axis.
C. Connect Pivot and Z to find MULTIPLIER.

FINAL CALCULATION: $L = L(lin) \cdot M$